

Derivative and Integral — Topics and Subtopics

503 subtopics — THE DERIVATIVE 280 · THE INTEGRAL 213 · THE CONNECTION 10

PART A — THE DERIVATIVE

A.1 Limits and continuity

- A.1.1 Increment of the argument
- A.1.2 Increment of the function
- A.1.3 Difference quotient
- A.1.4 Average rate of change
- A.1.5 Secant line
- A.1.6 Tangent as the limiting position of a secant
- A.1.7 Neighbourhood of a point
- A.1.8 Limit of a sequence
- A.1.9 Limit of a function at a point
- A.1.10 Formal definition of a limit (ϵ - δ)
- A.1.11 One-sided limits
- A.1.12 Limit laws — sum, difference, product, quotient
- A.1.13 Limit of a composite function
- A.1.14 Limit of a power and a root
- A.1.15 Squeeze theorem
- A.1.16 Factorising and cancelling
- A.1.17 Conjugate expressions
- A.1.18 Dividing by the highest power
- A.1.19 Change of variable in a limit
- A.1.20 Indeterminate form $0/0$
- A.1.21 Indeterminate form ∞/∞
- A.1.22 Indeterminate form $\infty - \infty$
- A.1.23 Indeterminate form $0 \cdot \infty$
- A.1.24 Indeterminate forms 1^∞ , 0^0 , ∞^0
- A.1.25 Standard limit $\sin x / x \rightarrow 1$
- A.1.26 Standard limit $\tan x / x \rightarrow 1$
- A.1.27 Standard limit $(1 - \cos x)/x^2 \rightarrow 1/2$
- A.1.28 Standard limit $(1 + 1/n)^n \rightarrow e$
- A.1.29 Standard limit $(1 + x)^{1/x} \rightarrow e$
- A.1.30 Standard limit $(e^x - 1)/x \rightarrow 1$
- A.1.31 Standard limit $\ln(1 + x)/x \rightarrow 1$
- A.1.32 Standard limit $(a^x - 1)/x \rightarrow \ln a$
- A.1.33 Limits at infinity
- A.1.34 Infinite limits
- A.1.35 Horizontal asymptotes
- A.1.36 Vertical asymptotes
- A.1.37 Oblique asymptotes
- A.1.38 Growth hierarchy of x^n , e^x , $\ln x$
- A.1.39 Continuity at a point
- A.1.40 Continuity on an interval
- A.1.41 One-sided continuity
- A.1.42 Operations on continuous functions
- A.1.43 Removable discontinuity
- A.1.44 Jump discontinuity
- A.1.45 Infinite discontinuity
- A.1.46 Oscillating discontinuity

- A.1.47 Intermediate value theorem
- A.1.48 Boundedness theorem
- A.1.49 Extreme value theorem

A.2 Definition of the derivative

- A.2.1 Limit of the difference quotient
- A.2.2 Derivative at a point
- A.2.3 Derivative as a function
- A.2.4 Notation $f'(x)$
- A.2.5 Notation dy/dx
- A.2.6 Notation y'
- A.2.7 Notation d/dx
- A.2.8 Notation $\dot{y}$
- A.2.9 Differentiation from first principles
- A.2.10 First principles for a constant
- A.2.11 First principles for x , x^2 , x^3 , x^n
- A.2.12 First principles for $1/x$ and $1/x^2$
- A.2.13 First principles for $\sqrt{x}$
- A.2.14 First principles for a quadratic
- A.2.15 First principles for $\sin x$ and $\cos x$
- A.2.16 First principles for e^x and $\ln x$
- A.2.17 One-sided derivatives
- A.2.18 Differentiability at a point
- A.2.19 Differentiability on an interval
- A.2.20 Differentiable implies continuous
- A.2.21 Continuous but not differentiable
- A.2.22 Corner point
- A.2.23 Cusp
- A.2.24 Vertical tangent
- A.2.25 Non-differentiability at a discontinuity
- A.2.26 Non-differentiability at an endpoint

A.3 Meaning of the derivative

- A.3.1 Gradient of the tangent
- A.3.2 $f'(a) = \tan \alpha$
- A.3.3 Sign of the derivative and the direction of the curve
- A.3.4 Instantaneous velocity
- A.3.5 Acceleration
- A.3.6 Instant of rest
- A.3.7 Jerk
- A.3.8 General rate of change
- A.3.9 Units of a derivative
- A.3.10 Marginal cost
- A.3.11 Marginal revenue
- A.3.12 Marginal profit
- A.3.13 Elasticity of demand

- A.3.14 Rate of a chemical reaction
- A.3.15 Rate of population growth
- A.3.16 Rate of cooling
- A.3.17 Current as the rate of change of charge

A.4 Table of derivatives

- A.4.1 Derivative of a constant
- A.4.2 Derivative of x
- A.4.3 Power rule for x^n
- A.4.4 Power rule for a natural index
- A.4.5 Power rule for an integer index
- A.4.6 Power rule for a fractional index
- A.4.7 Power rule for a real index
- A.4.8 Derivative of $1/x$
- A.4.9 Derivative of $1/x^n$
- A.4.10 Derivative of $\sqrt{x}$
- A.4.11 Derivative of $\sqrt[n]{x}$
- A.4.12 Derivative of $\sin x$
- A.4.13 Derivative of $\cos x$
- A.4.14 Derivative of $\tan x$
- A.4.15 Derivative of $\cot x$
- A.4.16 Derivative of $\sec x$
- A.4.17 Derivative of $\operatorname{cosec} x$
- A.4.18 Derivative of e^x
- A.4.19 Derivative of a^x
- A.4.20 Derivative of $\ln x$
- A.4.21 Derivative of $\ln|x|$
- A.4.22 Derivative of $\log_a x$
- A.4.23 Derivative of $\arcsin x$
- A.4.24 Derivative of $\arccos x$
- A.4.25 Derivative of $\arctan x$
- A.4.26 Derivative of $\operatorname{arccot} x$
- A.4.27 Derivative of $\sinh x$, $\cosh x$, $\tanh x$

A.5 Rules of differentiation

- A.5.1 Constant multiple rule
- A.5.2 Sum rule
- A.5.3 Difference rule
- A.5.4 Product rule
- A.5.5 Product rule for three factors
- A.5.6 Quotient rule
- A.5.7 Reciprocal rule
- A.5.8 Chain rule
- A.5.9 Inner and outer function
- A.5.10 Chain rule for $(ax + b)^n$
- A.5.11 Chains of three or more links
- A.5.12 Chain rule combined with the product rule
- A.5.13 Chain rule combined with the quotient rule
- A.5.14 Derivative of an inverse function
- A.5.15 Simplifying before differentiating

A.6 Techniques of differentiation

- A.6.1 Implicit differentiation
- A.6.2 Implicit differentiation of y^n
- A.6.3 Implicit differentiation of a product xy
- A.6.4 Tangents to the circle and the ellipse
- A.6.5 Logarithmic differentiation
- A.6.6 Differentiating $f(x)^{g(x)}$
- A.6.7 Differentiating $x^{f(x)}$

- A.6.8 Parametric differentiation
- A.6.9 Second derivative of a parametric curve
- A.6.10 Differentiating a piecewise function
- A.6.11 Differentiating a modulus function
- A.6.12 Differentials $dy = f'(x)dx$

A.7 Higher-order derivatives

- A.7.1 Second derivative
- A.7.2 Third derivative
- A.7.3 n -th derivative
- A.7.4 Notation d^2y/dx^2 , $f^{(n)}(x)$
- A.7.5 Second derivative as the rate of change of the gradient
- A.7.6 Patterns in $f^{(n)}$ for e^{ax} , $\sin x$, $1/x$, $\ln x$
- A.7.7 Leibniz rule for the n -th derivative of a product

A.8 Tangent and normal

- A.8.1 Gradient of the tangent at a point
- A.8.2 Point-gradient form of the tangent
- A.8.3 Tangent at a given x
- A.8.4 Tangent at a given y
- A.8.5 Tangent with a given gradient
- A.8.6 Horizontal tangents
- A.8.7 Vertical tangents
- A.8.8 Tangent from an external point
- A.8.9 Gradient of the normal
- A.8.10 Equation of the normal
- A.8.11 Axis intercepts of a tangent or normal
- A.8.12 Area of the triangle cut by a tangent
- A.8.13 Subtangent and subnormal
- A.8.14 Where a tangent meets the curve again
- A.8.15 Common tangent to two curves
- A.8.16 Angle between two curves
- A.8.17 Orthogonal curves

A.9 Behaviour of a function

- A.9.1 Increasing function and $f' > 0$
- A.9.2 Decreasing function and $f' < 0$
- A.9.3 Constant function and $f' = 0$
- A.9.4 Intervals of monotonicity
- A.9.5 Sign chart
- A.9.6 Proving an inequality by monotonicity
- A.9.7 Stationary point
- A.9.8 Critical point
- A.9.9 Fermat's theorem
- A.9.10 First derivative test
- A.9.11 Second derivative test
- A.9.12 Stationary point of inflection
- A.9.13 Local maximum
- A.9.14 Local minimum
- A.9.15 Global maximum and minimum
- A.9.16 Closed-interval method
- A.9.17 Greatest and least value on $[a, b]$
- A.9.18 Concave up
- A.9.19 Concave down
- A.9.20 Point of inflection
- A.9.21 Non-stationary inflection
- A.9.22 Curve-sketching algorithm
- A.9.23 Sketching a polynomial

- A.9.24 Sketching a rational function
- A.9.25 Sketching a function with a radical
- A.9.26 Sketching an exponential or logarithmic function
- A.9.27 Sketching a trigonometric function
- A.9.28 Sketching a function with a modulus
- A.9.29 Sketching f' from the graph of f
- A.9.30 Deducing f from the graph of f'
- A.9.31 Matching f , f' and f''

A.10 Mean value theorems

- A.10.1 Rolle's theorem
- A.10.2 Lagrange's mean value theorem
- A.10.3 Cauchy's mean value theorem
- A.10.4 Constant-function corollary
- A.10.5 Equal-derivatives corollary
- A.10.6 L'Hôpital's rule
- A.10.7 Repeated application of L'Hôpital's rule
- A.10.8 Converting other indeterminate forms to a quotient

A.11 Optimisation

- A.11.1 Objective function
- A.11.2 Constraint
- A.11.3 Elimination of a variable
- A.11.4 Feasible domain
- A.11.5 Justification of the nature of an extremum
- A.11.6 Largest rectangle in a given perimeter
- A.11.7 Largest rectangle in a semicircle
- A.11.8 Largest rectangle under a parabola
- A.11.9 Box of greatest volume from a cut sheet
- A.11.10 Cylinder of least surface area
- A.11.11 Cone of greatest volume in a sphere
- A.11.12 Shortest distance from a point to a curve
- A.11.13 Largest triangle in a circle
- A.11.14 Least perimeter for a fixed area
- A.11.15 Ladder round a corner
- A.11.16 Trough of greatest capacity
- A.11.17 Greatest height and range of a projectile
- A.11.18 Least time and least energy paths
- A.11.19 Maximum power transfer
- A.11.20 Maximum profit
- A.11.21 Minimum average cost
- A.11.22 Optimal price
- A.11.23 Optimal order quantity

A.12 Rates of change and kinematics

- A.12.1 Connected rates of change
- A.12.2 Chain rule for related rates
- A.12.3 Expanding circle and sphere

- A.12.4 Draining cone
- A.12.5 Rising water level
- A.12.6 Sliding ladder
- A.12.7 Lengthening shadow
- A.12.8 Inflating balloon
- A.12.9 Spreading oil slick
- A.12.10 Displacement, velocity, acceleration
- A.12.11 Initial conditions at $t = 0$
- A.12.12 Change of direction
- A.12.13 Displacement against distance travelled
- A.12.14 Maximum displacement and maximum speed
- A.12.15 Motion under variable acceleration
- A.12.16 Displacement-time graphs
- A.12.17 Velocity-time graphs
- A.12.18 The modelling cycle
- A.12.19 Interpreting the sign and size of a rate
- A.12.20 Limitations of a model

A.13 Approximation

- A.13.1 Small increments formula
- A.13.2 Linear approximation
- A.13.3 Tangent line approximation
- A.13.4 Choice of a convenient point
- A.13.5 Absolute error
- A.13.6 Relative error
- A.13.7 Percentage error
- A.13.8 Propagation of error in area and volume
- A.13.9 Newton-Raphson iteration
- A.13.10 Locating a root by sign change
- A.13.11 Choice of a starting value
- A.13.12 Failure of Newton-Raphson
- A.13.13 Maclaurin series
- A.13.14 Taylor series
- A.13.15 Series for e^x
- A.13.16 Series for $\sin x$ and $\cos x$
- A.13.17 Series for $\ln(1 + x)$
- A.13.18 Binomial series for $(1 + x)^n$
- A.13.19 Interval of validity
- A.13.20 Evaluating a limit by a series

A.14 The modulus function

- A.14.1 Graph of $y = |x|$
- A.14.2 Graph of $y = |ax + b|$
- A.14.3 Graph of $y = |f(x)|$
- A.14.4 Graph of $y = f(|x|)$
- A.14.5 Non-differentiability of $|x|$ at 0
- A.14.6 Derivative of $|x|$ as $\operatorname{sgn} x$
- A.14.7 Solving $|f(x)| = g(x)$
- A.14.8 Solving $|f(x)| < g(x)$

PART B — THE INTEGRAL

B.1 The antiderivative

- B.1.1 Definition of an antiderivative
- B.1.2 Checking an antiderivative by differentiation
- B.1.3 The family $F(x) + C$
- B.1.4 Uniqueness up to a constant
- B.1.5 Family of parallel curves
- B.1.6 Initial condition
- B.1.7 Particular antiderivative
- B.1.8 Antiderivative through a given point

B.2 The indefinite integral

- B.2.1 Notation $\int f(x) dx$
- B.2.2 Integrand
- B.2.3 Variable of integration
- B.2.4 Constant of integration
- B.2.5 Integration as the inverse of differentiation
- B.2.6 Constant multiple rule
- B.2.7 Sum and difference rule
- B.2.8 Term-by-term integration
- B.2.9 Absence of a product rule for integration
- B.2.10 Absence of a quotient rule for integration

B.3 Table of integrals

- B.3.1 $\int x^n dx, n \neq -1$
- B.3.2 $\int dx$
- B.3.3 $\int dx/x = \ln|x| + C$
- B.3.4 The exceptional case $n = -1$
- B.3.5 $\int dx/x^2$
- B.3.6 $\int \sqrt{x} dx$
- B.3.7 $\int e^x dx$
- B.3.8 $\int e^{(ax+b)} dx$
- B.3.9 $\int a^x dx$
- B.3.10 $\int \ln x dx$
- B.3.11 $\int \sin x dx$
- B.3.12 $\int \cos x dx$
- B.3.13 $\int dx/\cos^2 x$
- B.3.14 $\int dx/\sin^2 x$
- B.3.15 $\int \tan x dx$
- B.3.16 $\int \cot x dx$
- B.3.17 $\int \sec x dx$
- B.3.18 $\int \operatorname{cosec} x dx$
- B.3.19 $\int dx/(x^2 + a^2)$
- B.3.20 $\int dx/(a^2 - x^2)$
- B.3.21 $\int dx/(x^2 - a^2)$
- B.3.22 $\int dx/\sqrt{a^2 - x^2}$
- B.3.23 $\int dx/\sqrt{x^2 + a^2}$
- B.3.24 $\int dx/\sqrt{x^2 - a^2}$
- B.3.25 $\int \sinh x dx$ and $\int \cosh x dx$

B.4 Techniques of integration

- B.4.1 Expanding a product first
- B.4.2 Splitting a fraction term by term
- B.4.3 Rewriting a surd or reciprocal as a power
- B.4.4 Using a trigonometric identity

- B.4.5 Dividing an improper algebraic fraction
- B.4.6 Completing the square
- B.4.7 Linear substitution for $(ax + b)^n$
- B.4.8 Linear substitution for $1/(ax + b)$
- B.4.9 Linear substitution in a trigonometric integral
- B.4.10 Substitution $u = g(x)$
- B.4.11 Changing dx to du
- B.4.12 Recognising the derivative of the inner function
- B.4.13 $\int f'(x)/f(x) dx$
- B.4.14 $\int f'(x) \cdot f(x)^n dx$
- B.4.15 $\int f'(x) \cdot e^{f(x)} dx$
- B.4.16 Substitution in a definite integral
- B.4.17 Changing the limits of integration
- B.4.18 Integration by parts
- B.4.19 LIATE choice of u
- B.4.20 Integration by parts applied twice
- B.4.21 Cyclic integration by parts
- B.4.22 Reduction formulae
- B.4.23 Definite integration by parts
- B.4.24 Partial fractions with distinct linear factors
- B.4.25 Partial fractions with a repeated linear factor
- B.4.26 Partial fractions with an irreducible quadratic factor
- B.4.27 Comparing coefficients
- B.4.28 Substituting convenient values
- B.4.29 Cover-up method
- B.4.30 Odd powers of $\sin$ and $\cos$
- B.4.31 Even powers of $\sin$ and $\cos$
- B.4.32 Products $\sin mx \cdot \cos nx$
- B.4.33 Powers of $\tan$ and $\sec$
- B.4.34 Weierstrass substitution $t = \tan(x/2)$
- B.4.35 Trigonometric substitution $x = a \sin \theta$
- B.4.36 Trigonometric substitution $x = a \tan \theta$
- B.4.37 Trigonometric substitution $x = a \sec \theta$
- B.4.38 Rationalising substitution $u = \sqrt[n]{ax + b}$
- B.4.39 Substitution $u = \sqrt{x}$
- B.4.40 Choosing a technique
- B.4.41 Checking an integral by differentiation
- B.4.42 Integrals with no elementary antiderivative

B.5 The definite integral

- B.5.1 The area problem
- B.5.2 Partition of an interval
- B.5.3 Lower (inscribed) sum
- B.5.4 Upper (circumscribed) sum
- B.5.5 Midpoint sum
- B.5.6 Integral (Riemann) sum
- B.5.7 Definition as a limit of sums
- B.5.8 Evaluating an integral from the definition
- B.5.9 Sum formulae Σi and Σi^2
- B.5.10 Integrability
- B.5.11 Notation and the limits of integration
- B.5.12 Lower limit and upper limit
- B.5.13 The dummy variable
- B.5.14 Definite integral as a number

- B.5.15 Zero-width property
- B.5.16 Reversal of the limits
- B.5.17 Linearity
- B.5.18 Additivity over adjacent intervals
- B.5.19 Comparison property
- B.5.20 Bounds on an integral
- B.5.21 Modulus inequality for integrals
- B.5.22 Even and odd symmetry
- B.5.23 Periodic integrands
- B.5.24 Integral as a function of its upper limit
- B.5.25 Fundamental theorem, part 1
- B.5.26 Differentiating $\int(a \text{ to } g(x)) f(t)dt$
- B.5.27 Newton–Leibniz formula
- B.5.28 Cancellation of the constant of integration
- B.5.29 Evaluating with square-bracket notation
- B.5.30 Conditions for the Newton–Leibniz formula
- B.5.31 Mean value of a function
- B.5.32 Mean value theorem for integrals
- B.5.33 Root mean square value

B.6 Applications of the definite integral

- B.6.1 Area under a curve above the axis
- B.6.2 Area under a curve below the axis
- B.6.3 A curve crossing the axis
- B.6.4 Signed area against total area
- B.6.5 Area between a curve and the y-axis
- B.6.6 Finding a limit from a given area
- B.6.7 Area between two curves
- B.6.8 Finding the points of intersection
- B.6.9 Identifying the upper and lower curve
- B.6.10 Curves that swap over
- B.6.11 Area with respect to y
- B.6.12 Region bounded by three curves
- B.6.13 Area bounded by a curve and its tangent
- B.6.14 Volume of revolution about the x-axis
- B.6.15 Volume of revolution about the y-axis
- B.6.16 Washer (annulus) method
- B.6.17 Volume between two surfaces of revolution
- B.6.18 Deriving the volume of a cone, sphere and paraboloid
- B.6.19 Rotation about a line $y = k$ or $x = k$
- B.6.20 Shell method
- B.6.21 Arc length
- B.6.22 Surface area of a solid of revolution
- B.6.23 Centroid of a plane region
- B.6.24 Centre of mass
- B.6.25 Work done by a variable force
- B.6.26 Work to empty a tank
- B.6.27 Hydrostatic pressure
- B.6.28 Moment of inertia
- B.6.29 Velocity from acceleration
- B.6.30 Displacement from velocity
- B.6.31 Distance travelled as $\int|v| dt$
- B.6.32 Area under a velocity–time graph
- B.6.33 Total from a rate
- B.6.34 Total charge from a current
- B.6.35 Probability from a density function
- B.6.36 Expectation and variance from a density function

B.7 Numerical integration

- B.7.1 Need for a numerical method
- B.7.2 Integration of tabulated data
- B.7.3 Trapezium rule
- B.7.4 Strips and ordinates
- B.7.5 Strip width h
- B.7.6 Tabulating the ordinates
- B.7.7 Over-estimate and under-estimate
- B.7.8 Concavity and the direction of the error
- B.7.9 Improving the estimate by doubling the strips
- B.7.10 Order of the error
- B.7.11 Simpson's rule
- B.7.12 Mid-ordinate rule
- B.7.13 Rectangle (endpoint) sums
- B.7.14 Monte Carlo estimation

B.8 Improper integrals

- B.8.1 Infinite limit of integration
- B.8.2 Unbounded integrand
- B.8.3 Convergence and divergence
- B.8.4 $\int(1 \text{ to } \infty) dx/x^p$
- B.8.5 $\int(0 \text{ to } \infty) e^{(-x)} dx$
- B.8.6 Integrals through a discontinuity
- B.8.7 The Gaussian integral

B.9 Differential equations

- B.9.1 Definition of a differential equation
- B.9.2 Order and degree
- B.9.3 General solution
- B.9.4 Particular solution
- B.9.5 Arbitrary constant
- B.9.6 Family of solution curves
- B.9.7 Verifying a solution
- B.9.8 Forming an equation from a rate statement
- B.9.9 Rate proportional to y
- B.9.10 Rate proportional to a power of t
- B.9.11 Rate inversely proportional to y
- B.9.12 Rate proportional to a difference
- B.9.13 Sign of the constant of proportionality
- B.9.14 $dy/dx = f(x)$
- B.9.15 $dy/dx = g(y)$
- B.9.16 Separable equations
- B.9.17 Separation of variables
- B.9.18 Applying an initial condition
- B.9.19 Making y the subject
- B.9.20 Absorbing a constant after a logarithm
- B.9.21 Exponential growth
- B.9.22 Exponential decay
- B.9.23 Half-life and doubling time
- B.9.24 Radioactive decay
- B.9.25 Compound interest
- B.9.26 Population growth
- B.9.27 Newton's law of cooling
- B.9.28 Mixing and dilution
- B.9.29 Draining tank (Torricelli)
- B.9.30 Logistic model
- B.9.31 Carrying capacity
- B.9.32 Long-run behaviour
- B.9.33 Direction fields

B.9.34 Equilibrium solutions and stability
B.9.35 Euler's method
B.9.36 First-order linear equations

B.9.37 Integrating factor
B.9.38 Second-order equations with constant coefficients

PART C — THE CONNECTION

C.1	Differentiation and integration as inverse operations	C.6	s, v, a in both directions
C.2	Fundamental theorem of calculus, part 1	C.7	Marginal quantity against total quantity
C.3	Fundamental theorem of calculus, part 2	C.8	Density against mass
C.4	Gradient against area	C.9	Net change as the integral of a rate
C.5	Instantaneous rate against accumulated total	C.10	Checking an integral by differentiating